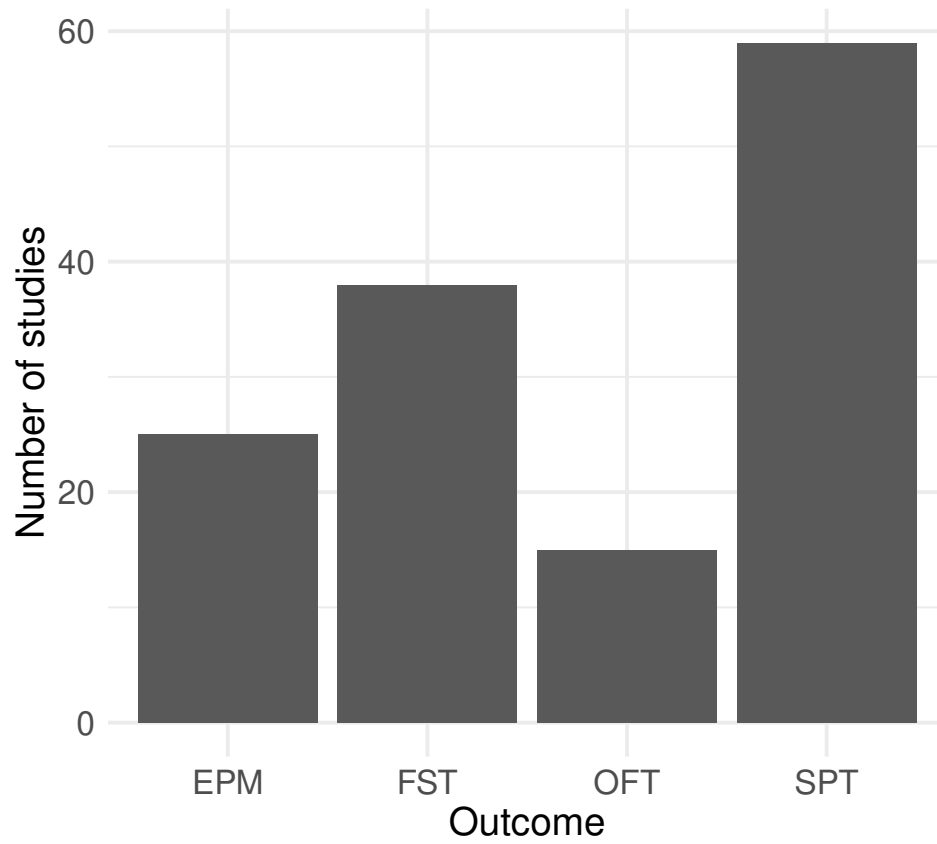


CRS studies meta analysis

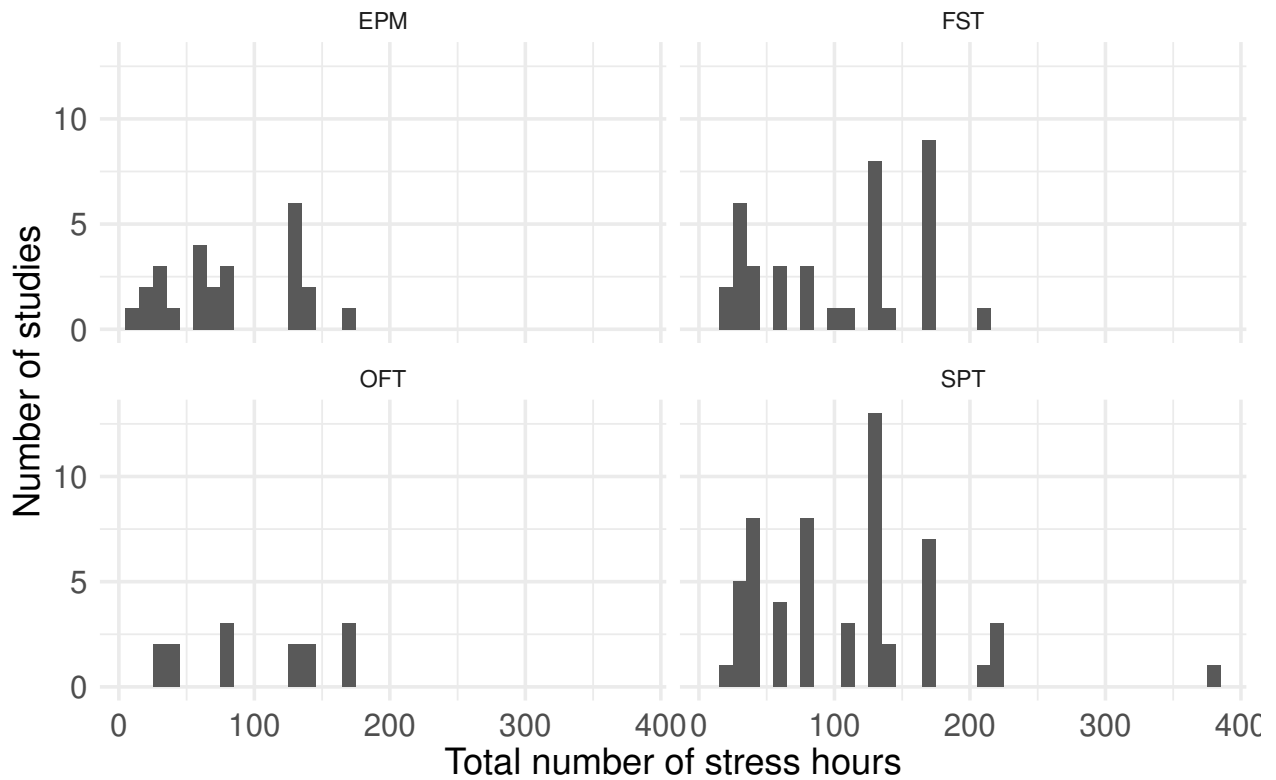
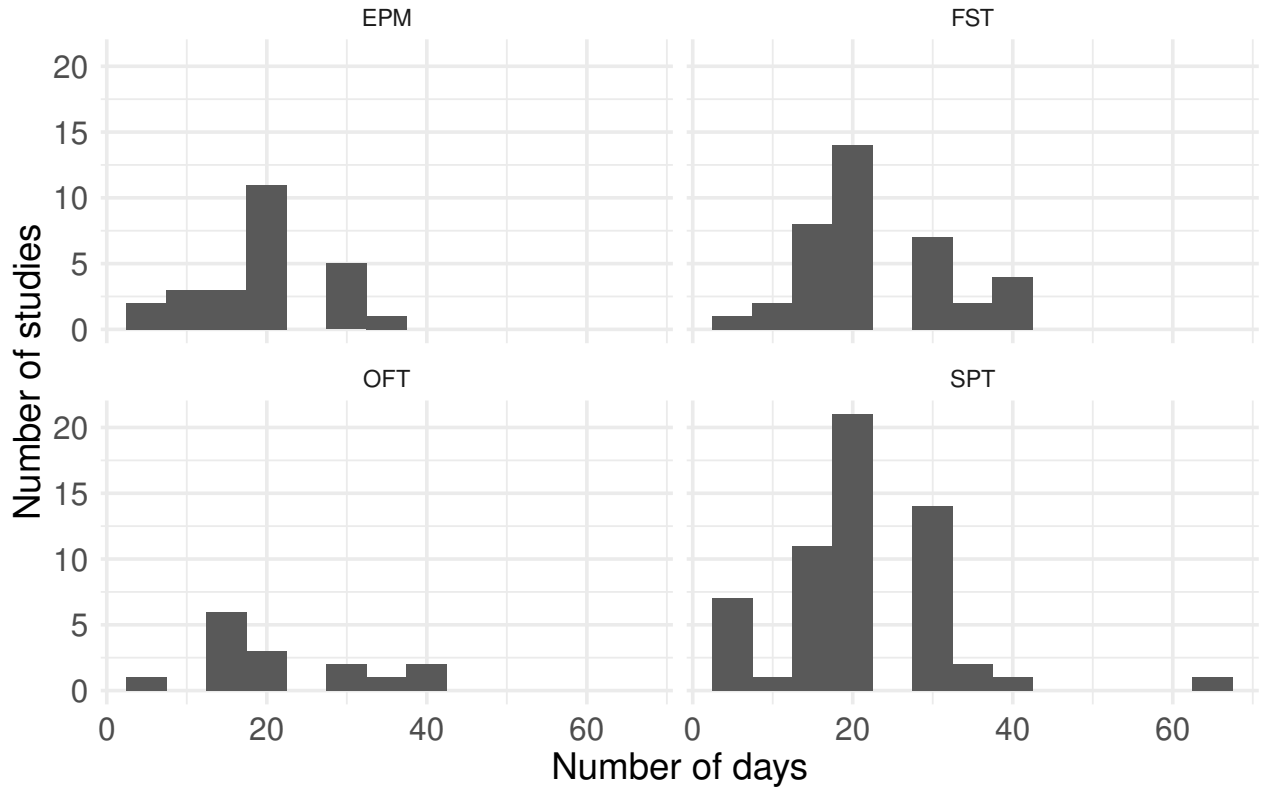
Last updated 2025-08-07

Data exploration

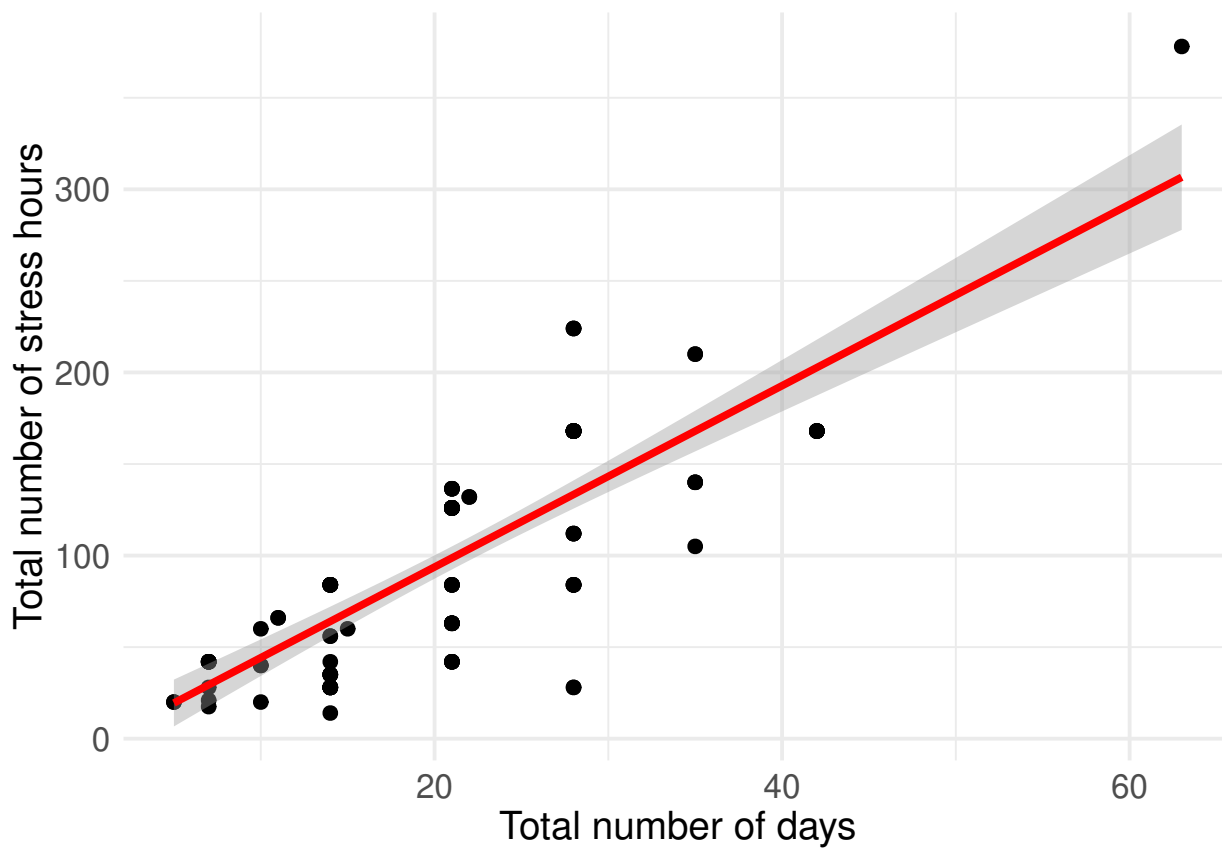
Number of studies



Length of stress

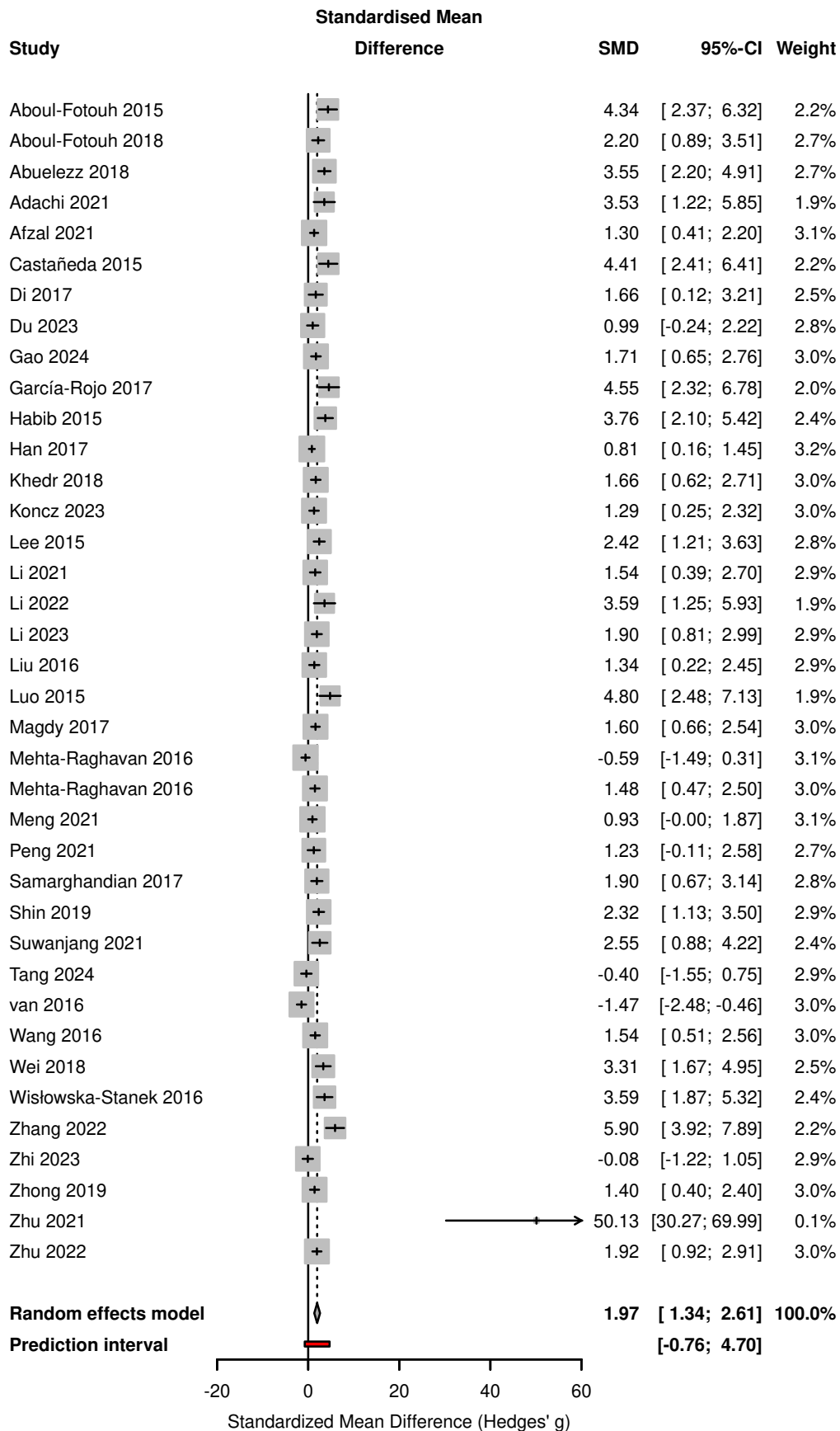


The number of days of chronic stress is highly collinear with the total time ($\rho = 0.78$), so we will only use total time for the analysis.



Below we will look at the effect of CSR on the different outcomes.

FST



There is a positive and statistically significant effect of CSR on the time spent immobile ($p = 2.46e-07$). The standardized mean difference (Hedges' g) is 1.97, roughly equivalent to a predicted raw mean difference of 40.75 seconds (the raw mean difference of the studies from the data is 45.19 seconds). The heterogeneity is very high ($I^2 = 80.8\%$), which supports the use of a random effects model.

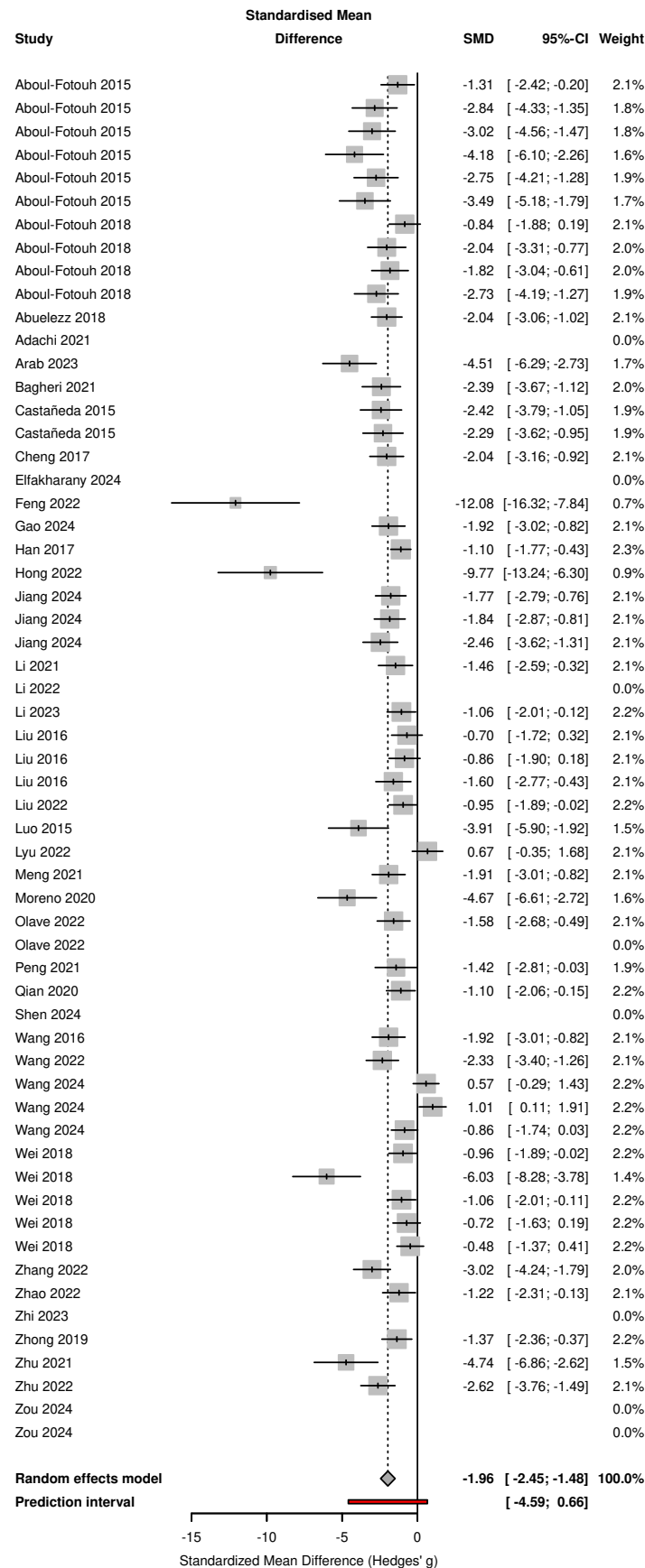
We can now ask the question of whether the total stress time is associated with the effect size.

In the case of FST I looked at total number of days stress was applied for.

```
##
## Mixed-Effects Model (k = 38; tau^2 estimator: REML)
##
##   logLik  deviance      AIC      BIC      AICc
## -78.0951  156.1903  162.1903  166.9408  162.9403
##
## tau^2 (estimated amount of residual heterogeneity):      1.5516 (SE = 0.4813)
## tau (square root of estimated tau^2 value):             1.2456
## I^2 (residual heterogeneity / unaccounted variability): 80.29%
## H^2 (unaccounted variability / sampling variability):    5.07
## R^2 (amount of heterogeneity accounted for):             11.47%
##
## Test for Residual Heterogeneity:
## QE(df = 36) = 168.7850, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 36) = 2.0002, p-val = 0.1659
##
## Model Results:
##
##              estimate      se    tval  df    pval    ci.lb    ci.ub
## intrcpt          0.9174  0.7930  1.1569  36  0.2549   -0.6909   2.5256
## total_days        0.0458  0.0324  1.4143  36  0.1659   -0.0199   0.1115
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Total stress time is **not** significantly associated with the effect size ($p = 0.17$). Indeed residual heterogeneity is still very high ($I^2 = 80.3\%$), meaning that the model with the added total time covariate still does not explain the heterogeneity in the data.

SPT

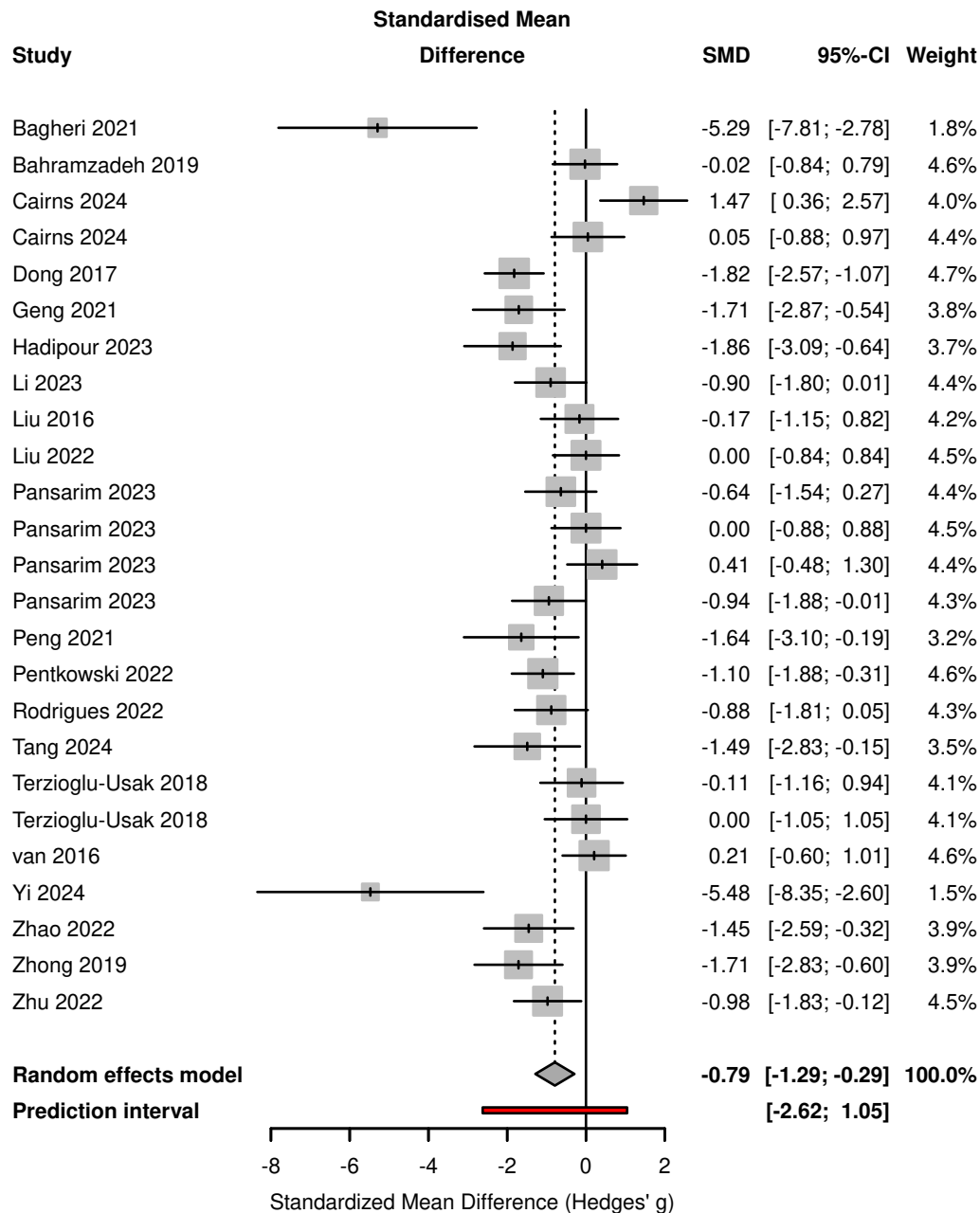


For the SPT, we have a negative and statistically significant effect of CSR on the % preference for sucrose ($p = 8.76e-11$). The standardized mean difference (Hedges' g) is -1.96, roughly equivalent to a predicted raw mean difference of -19.62% (the raw mean difference of the studies from the data is -20.12%). Heterogeneity is very high in this case as well ($I^2 = 79.1\%$).

```
##
## Mixed-Effects Model (k = 43; tau^2 estimator: REML)
##
##   logLik  deviance      AIC      BIC      AICc
## -74.6988  149.3976  157.3976  164.1532  158.5405
##
## tau^2 (estimated amount of residual heterogeneity):      1.4230 (SE = 0.4050)
## tau (square root of estimated tau^2 value):             1.1929
## I^2 (residual heterogeneity / unaccounted variability): 81.42%
## H^2 (unaccounted variability / sampling variability):    5.38
## R^2 (amount of heterogeneity accounted for):             1.21%
##
## Test for Residual Heterogeneity:
## QE(df = 40) = 180.7324, p-val < .0001
##
## Test of Moderators (coefficients 2:3):
## F(df1 = 2, df2 = 40) = 0.8319, p-val = 0.4426
##
## Model Results:
##
##              estimate      se      tval  df      pval      ci.lb      ci.ub
## intrcpt          -1.4837  0.5615  -2.6423  40  0.0117  -2.6185  -0.3488  *
## total_days        -0.0261  0.0220  -1.1852  40  0.2429  -0.0705   0.0184
## water_depriv_h     0.0156  0.0228   0.6814  40  0.4996  -0.0306   0.0617
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Regression analysis of the effect size against total stress time (in days), and the hours of water and food deprivation (note that these are the same, so they get combined into a single variable) shows that neither of these covariates is significantly associated with the effect size ($p = 0.24$ and 0.5 , respectively).

EPM



For the EPM we have a negative and statistically significant effect of CSR on the % time spent in the open arms ($p = 3.16e-03$). The SMD is -0.79, roughly equivalent to a predicted raw mean difference of -15.55% (the raw mean difference of the studies from the data is -13.8%). Heterogeneity is very high in this case as well ($I^2 = 72.7\%$).

```
##
## Mixed-Effects Model (k = 25; tau^2 estimator: REML)
##
##   logLik deviance      AIC      BIC    AICc
## -38.3405  76.6809  82.6809  86.0874  83.9441
##
## tau^2 (estimated amount of residual heterogeneity):    0.8084 (SE = 0.3253)
```

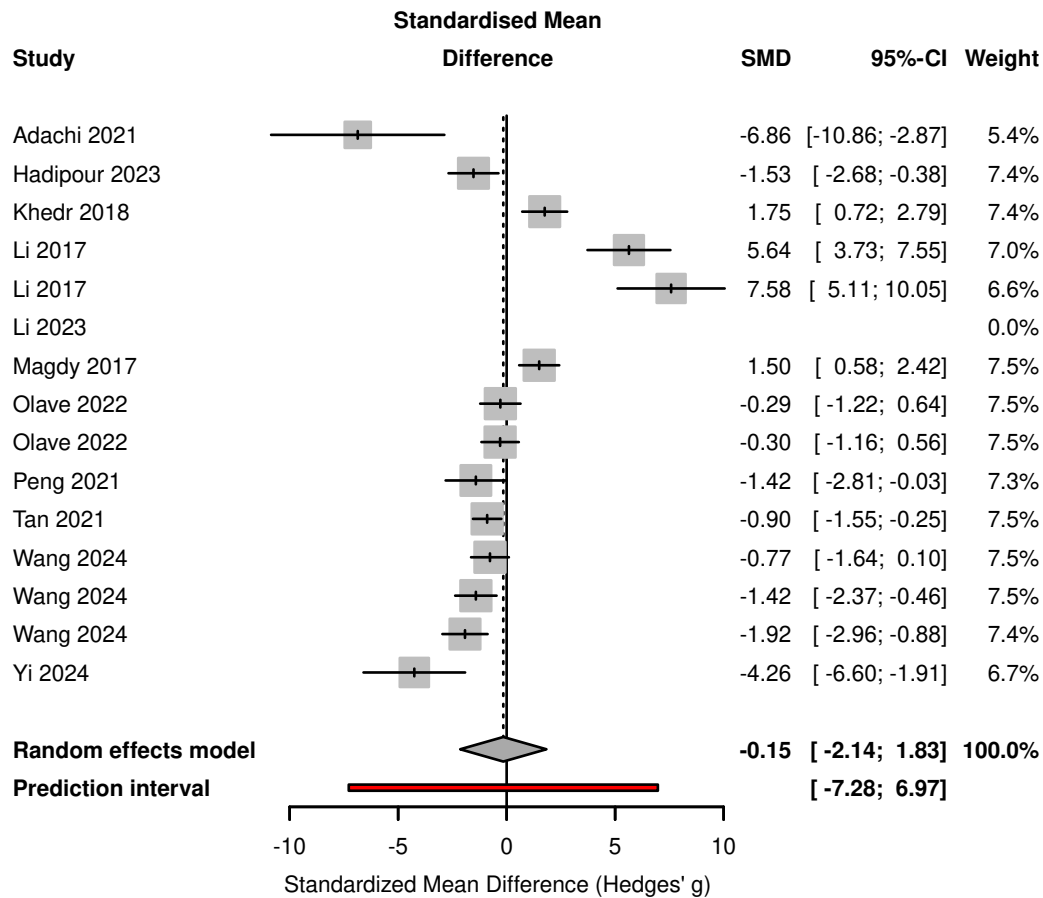
```

## tau (square root of estimated tau^2 value):          0.8991
## I^2 (residual heterogeneity / unaccounted variability): 75.85%
## H^2 (unaccounted variability / sampling variability):  4.14
## R^2 (amount of heterogeneity accounted for):          0.00%
##
## Test for Residual Heterogeneity:
## QE(df = 23) = 86.5250, p-val < .0001
##
## Test of Moderators (coefficient 2):
## F(df1 = 1, df2 = 23) = 0.3725, p-val = 0.5476
##
## Model Results:
##
##           estimate      se    tval  df    pval    ci.lb    ci.ub
## intrcpt      -0.4012  0.6920  -0.5797  23  0.5678   -1.8327   1.0304
## total_days   -0.0204  0.0334  -0.6103  23  0.5476   -0.0894   0.0487
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Regression analysis of the effect size against total stress time (in days) shows that it is not significantly associated with the effect size ($p = 0.55$).

OFT



For the OFT we do not have a statistically significant effect of CSR on the time spent in the center ($p = 0.87$). The SMD is -0.15, roughly equivalent to a predicted raw mean difference of -0.61 s (the raw mean difference of the studies from the data is -1.85 s). Heterogeneity is very high ($I^2 = 91.5\%$).

Notes on analysis method

1. The meta analysis has been done using the **meta** R package. Because of the large heterogeneity in the data, I used a random effects model for the meta-analysis. This assumes that not only the effect size of each study varies around the “true” effect size, but also that the true effect size itself varies across studies because e.g. of differences in study design.
2. Effect sizes were calculated as standardized mean differences (SMD) using Hedges’ g, which accounts for small sample bias. This is calculated as:

$$g = \frac{\bar{X}_e - \bar{X}_c}{s_p} * \left(1 - \frac{3}{4(n_e + n_c) - 9}\right)$$

where $\bar{X}_e$ and $\bar{X}_c$ are the means of the experimental and control groups, respectively, n_e and n_c are their sample sizes, and s_p is the pooled standard deviation, calculated as:

$$s_p = \sqrt{\frac{(n_e - 1)s_e^2 + (n_c - 1)s_c^2}{n_e + n_c - 2}}$$

A positive value for Hedges’ g indicates that the experimental group has a higher mean than the control group.

3. While there are rough guidelines on how to interpret Hedges’ g (e.g. 0.2 = small, 0.5 = medium, 0.8 = large), these have the same issues as Cohen’s d, such as being very context-dependent. One option would be to report the mean difference in the original units of measurement, which is more interpretable. However, this won’t consider all the corrections that the meta analysis does, so another option is to take the meta-analysis predicted Hedges’ g and convert it to a raw effect size in the original units of measurement by multiplying it by the average pooled standard deviation across all studies. In most cases the two will be very similar and will give a more intuitive interpretation of the effect size.
4. Some of the studies measured the same outcome separately in males and females. We consider those as separate studies; while this might slightly bias the results (because the studies are not completely independent), the number of such cases is small and the effect is likely negligible (?). I have not corrected for that. If we wanted to we could run a multi-level meta-analysis with study as a random effect.
5. Some studies do not report n; in these cases they will have a weight of 0% in the meta-analysis, which means they will not contribute to the overall effect size (we still show them in the forest plot).
6. The effect of covariates has been investigated using a meta-regression approach that measures the effect of a covariate (e.g. stress length) on the effect size.